The documentation and process conversion measures necessary to comply with this document shall be completed by 11 June 2021.

INCH-POUND

MIL-PRF-19500/596L
w/AMENDMENT 1
11 March 2021
SUPERSEDING
MIL-PRF-19500/596L
24 September 2019

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, FIELD EFFECT, N-CHANNEL, SILICON, REPETITIVE AVALANCHE,
TYPES 2N7218, 2N7219, 2N7221, 2N7222,
ENCAPSULATED (FLANGE AND SURFACE MOUNT PACKAGES) AND UNENCAPSULATED,
QUALITY LEVELS JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for N-channel, enhancement mode, MOSFET, power transistors. These transistors are intended for use in high density power switching applications and include ratings for avalanche energy (E_{AR} and E_{AS}) and avalanche current (I_{AR}). Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type and two levels of product assurance (JANHC and JANKC) are provided for un-encapsulated device types (die) as specified in MIL-PRF-19500.

1.2 Physical dimensions.

1.2.1 Package outlines. The device package outlines for the encapsulated device types are as follows: Three terminal flange mount package (TO-254AA) in accordance with figure 1 and three terminal surface mount device (SMD) package (TO-276AB) in accordance with figure 2.

1.2.2 Un-encapsulated die. The dimensions and topography for JANHC and JANKC un-encapsulated die are in accordance with figure 3.

1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Type	P_T (1) T_C = +25°C	P_T T_A = +25°C	$R_{\theta JC}$ (2)	$V_{(BR)DSS}$ min V_{GS} = 0 I_D = 1.0 mA dc	I_{D1} (3) (4) T_C = +25°C	I_{D2} (3) (4) T_C = +100°C	I_S	I_{DM} (5)	T_J and T_{STG}
	W	W	°C/W	V dc	A dc	A dc	A dc	A (pk)	°C
2N7218, 2N7218U	125	4	1.0	100	28	20	28	112	−55 to +150
2N7219, 2N7219U	125	4	1.0	200	18	11	18	72	−55 to +150
2N7221, 2N7221U	125	4	1.0	400	10	6	10	40	−55 to +150
2N7222, 2N7222U	125	4	1.0	500	8	5	8	32	−55 to +150

See notes next page.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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FSC 5961

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1.3 Maximum ratings - continued.

Type	I_{AR}	E_{AS}	E_{AR}	$V_{(ISO)}$ at 70,000 feet	V_{GS}	$r_{DS(on)\ max}$ (6) V_{GS} = 10 V dc, $I_D = I_{D2}$	
						T_J = +25°C	T_J = +150°C
	A	mj	mj		V dc	Ω	Ω
2N7218, 2N7218U	28	250	12.5		±20	0.077	0.154
2N7219, 2N7219U	18	450	12.5		±20	0.18	0.387
2N7221, 2N7221U	10	650	12.5	400	±20	0.55	1.32
2N7222, 2N7222U	8	700	12.5	500	±20	0.85	2.04

(1) Derate linearly 1.0 W/°C for T_C > +25°C.
(2) See figure 4, thermal impedance curves.
(3) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_J\ \text{max} - T_C}{(R_{\Theta JC}) x (R_{DS(ON)}\ at\ T_{Jmax})}}$$

(4) See figure 5, maximum drain current graph.
(5) I_{DM} = 4 x I_{D1} as calculated in note 3.
(6) Pulsed (see 4.5.1).

1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA dc		Max I_{DSS1} V_{GS} = 0, V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)1}$ (1) $I_D = I_{D2}$ V_{GS} = 10 V
	V dc	V dc		μA dc	Ohms
		Min	Max		
2N7218, 2N7218U	100	2.0	4.0	25	0.077
2N7219, 2N7219U	200	2.0	4.0	25	0.18
2N7221, 2N7221U	400	2.0	4.0	25	0.55
2N7222, 2N7222U	500	2.0	4.0	25	0.85

(1) Pulsed (see 4.5.1).

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction examples, 6.5 for a list of available PINs, and 6.6 for substitution information.

1.5.1 JAN certification mark and quality level.

1.5.1.1 Quality level designators for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV" and "JANS".

1.5.1.2 Quality level designators for un-encapsulated die. The quality level designators for un-encapsulated devices that are applicable for this specification sheet are "JANHC" and "JANKC".

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1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "2N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are "7218", "7219", "7221", and "7222".

1.5.3 Suffix symbols. The suffix symbols (or lack thereof) that designate the package outline for the devices covered by this specification sheet are as follows.

	A blank designator identifies that the package is a flange mount TO-254AA (see figure 1).
U	This designator identifies the package is a SMD TO-276AB (see figure 2).

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.5 Die identifiers for un-encapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet is "A" (see figure 3 and 6.5.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 – Semiconductor Devices, General Specification for.

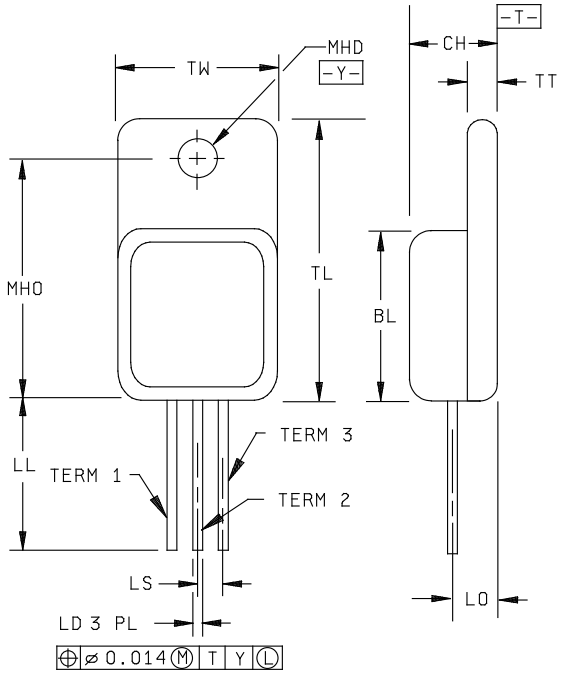
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 – Test Methods for Semiconductor Devices.
* MIL-STD-883 - Test Method Standard for Microcircuits.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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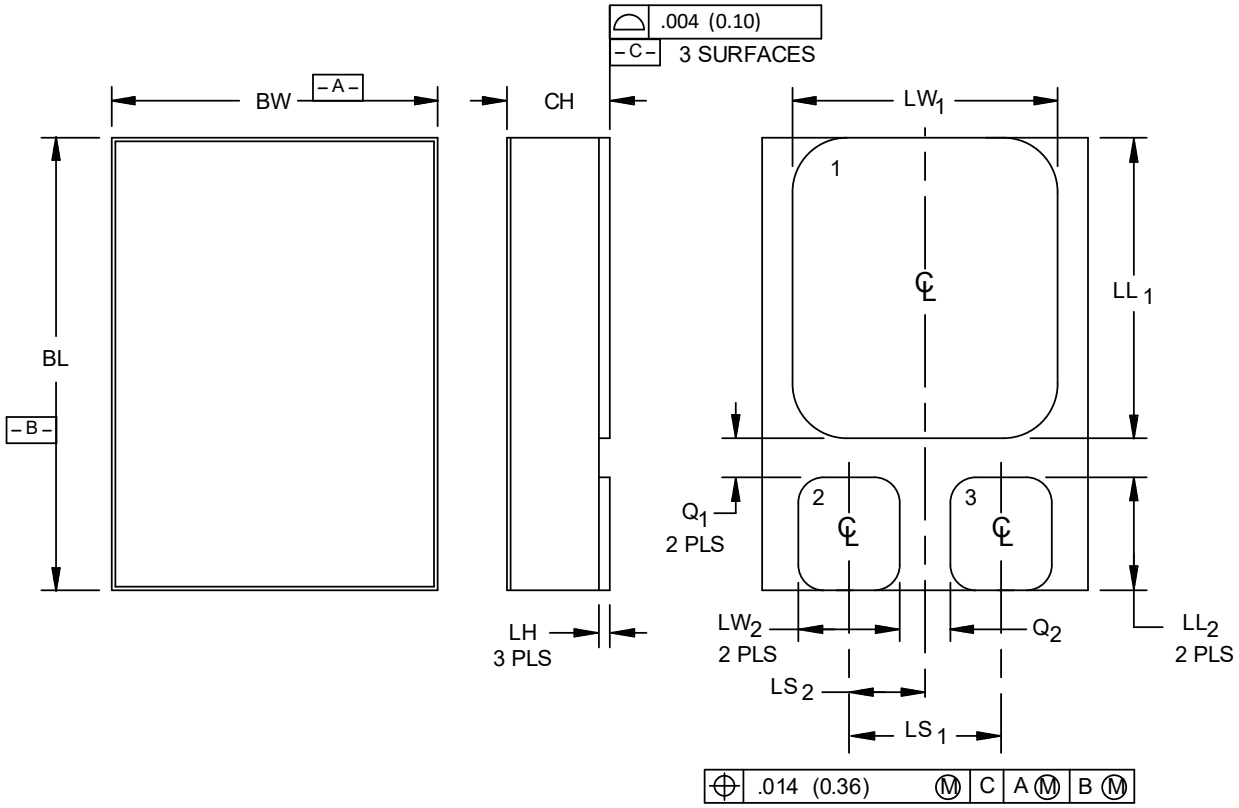


LTR	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.84
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.530	.550	13.46	13.97
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	3.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.84

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Term 1 is drain, term 2 is the source, and term 3 is the gate.
3. Protrusion thickness of ceramic eyelets included in dimension LL.
4. All terminals are isolated from case. Methods used for electrical isolation of the terminals feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic).
* 5. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for flange mount package (TO-254AA).

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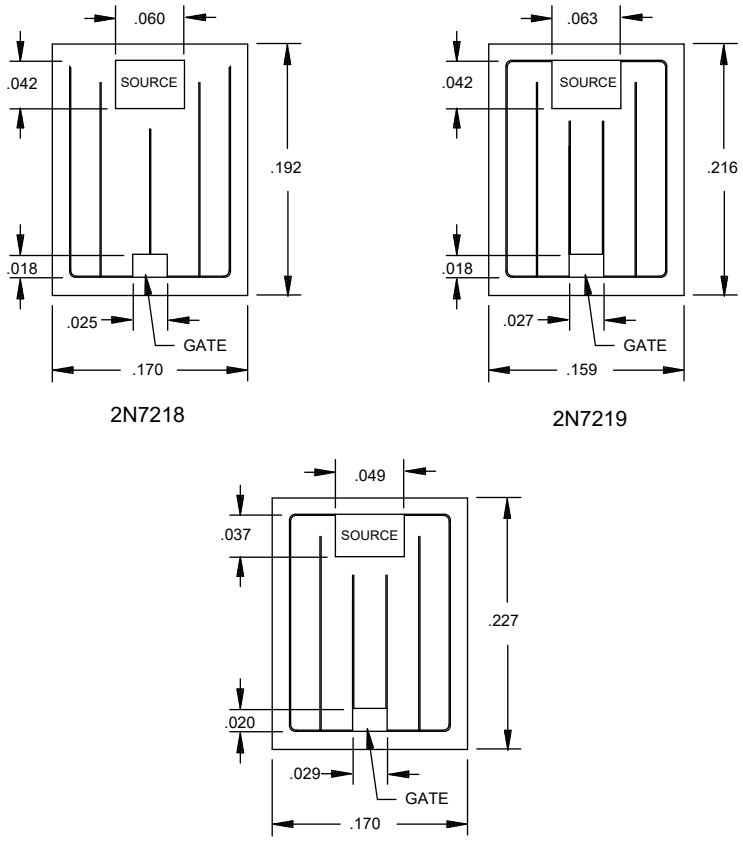


Dimensions				
Symbol	Inches		Millimeters	
	Min	Max	Min	Max
BL	.620	.630	15.75	16.00
BW	.445	.455	11.30	11.56
CH		.142		3.60
LH	.010	.020	0.26	0.50
LL_1	.410	.420	10.41	10.67
LL_2	.152	.162	3.86	4.11
LS_1	.210 BSC		5.33 BSC	
LS_2	.105 BSC		2.67 BSC	
LW_1	.370	.380	9.40	9.65
LW_2	.135	.145	3.43	3.68
Q_1	.030		0.76	
Q_2	.035		0.89	
Term 1	Drain			
Term 2	Gate			
Term 3	Source			

NOTES:
1. Dimensions are in inches. Millimeters are given for information only.
2. The lid shall be electrically isolated from the drain, gate and source.
* 3. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

FIGURE 2. Dimensions and configuration of SMD package (TO-276AB).

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Inches	mm	Inches	mm	Inches	mm
.018	0.46	.037	0.94	.159	4.04
.020	0.51	.042	1.07	.170	4.32
.025	0.64	.049	1.24	.192	4.88
.027	0.69	.060	1.52	.216	5.49
.029	0.74	.063	1.60	.227	5.77

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. The die thickness is .0187 inch (0.475 mm) for all types.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The metallization shall be as follows:
   a. The back metals are chromium, nickel, and silver.
      The thickness of the Cr-Ni-Ag is .75kÅ – 1.5kÅ – 1.87KÅ minimum.
   b. The top metal is aluminum. The thickness of the Al is 22.5kÅ minimum.

* 5. In accordance with ASME Y14.5, diameters are equivalent to ϕx symbology.

6. The back contact is the drain for all types.

* FIGURE 3. JANHC and JANKC (A-version) die dimensions.

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figures 1 (TO-254AA), 2 (TO-276AB), and 3 (die) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead formation or finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Lead formation. When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 of table E-IV of MIL-PRF-19500 and 100 percent dc testing in accordance with table I, subgroup 2 herein.

3.4.3 Lead isolation. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.4.4 Internal construction. Multiple chip construction shall not be permitted.

* 3.4.5 Silicone die coating. The use of a silicone die coat requires a successful completion of MIL-STD-883, method 5011 on each silicone lot for its intended applications, and as part of the full MIL-PRF-19500 qualification process.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500, except at the option of the manufacturer, the country of origin and/or the manufacturers identification may be omitted from the body of the transistor.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

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3.8 Electrostatic discharge sensitive (ESDS). The devices covered by this specification sheet have been classified as ESDS. The devices shall be handled in accordance with the ESD program established to comply with the requirements of MIL-PRF-19500 to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, R ≤ 100 kΩ, whenever bias voltage is to be applied drain to source.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

d. Element evaluation of un-encapsulated die (see 4.6).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with appendix G of MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (quality levels JANS, JANTX, and JANTXV only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (1) (2)	Measurement	
	JANS level	JANTX and JANTXV level
(3)	Gate stress test (see 4.3.1)	Gate stress test (see 4.3.1)
(3)	Method 3470 of MIL-STD-750. (see 4.3.2)	Method 3470 of MIL-STD-750. (see 4.3.2)
(3) 3c	Method 3161 of MIL-STD-750 (see 4.3.3)	Method 3161 of MIL-STD-750 (see 4.3.3)
* 5	Method 2052 of MIL-STD-750, PIND (see MIL-PRF-19500 and 4.3.5)	Not applicable
9	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, subgroup 2 of table I herein;	Subgroup 2 of table I herein
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table Iherein. ΔI_{GSSF1} = +20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = −20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ Subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A or T_A = +175°C and t = 48 hours
13	Subgroup 2 and 3 of table I herein. ΔI_{GSSF1} = +20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = −20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein. ΔI_{GSSF1} = +20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = −20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1}, and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Gate stress test. Apply V_{GS} = 30 V minimum for t = 250 µs minimum.

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4.3.2 Single pulse avalanche energy (E_{AS}). The test for single pulse avalanche energy shall be performed in accordance with method 3470 of MIL-STD-750. The following details shall apply:

a. Peak current (I_{AS}) ..... I_{D1}.

b. Peak gate voltage (V_{GS}) ..... 10 V.

c. Gate to source resistor (R_{GS}) ..... $25 \leq R_{GS} \leq 200\ \Omega$.

d. Initial case temperature ..... +25°C +10°C, −5°C.

e. Inductance: ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied ..... 1 pulse minimum.

g. Supply voltage (V_{DD}) ..... 50 V, 25 V for 100 V devices.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, and t_{MD} (and V_H where appropriate). See table II, group E, subgroup 4 herein.

4.3.4 Dielectric withstanding voltage. The dielectric withstanding voltage test shall be in accordance with test method 1081 of MIL-STD-750. The following details shall apply:

a. Magnitude of test voltage ..... 900 V dc.

b. Duration of application of test voltage ..... 15 seconds (minimum).

c. Points of application of test voltage ..... All leads to case (bunch connection).

d. Method of connection ..... Mechanical.

e. Kilovolt-ampere rating of high voltage source ..... 1200V/1.0 mA (minimum).

f. Maximum leakage current ..... 1.0 mA.

g. Voltage ramp up time ..... 500 V/second.

* 4.3.5 PIND. Not applicable in screening when devices are processed using alternative method and flow requirements approved by the qualifying activity, that includes incorporating the use of certified clean processing and silicone die coat. Instead, the PIND test performance shall be performed in group B3 and group C3, on a lot sample basis. PIND failures detected in group B or C will represent lot jeopardy and shall be evaluated for root cause and lot integrity.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and herein.

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4.4.2.1 Quality level JANS.

	Subgroup	Method	Condition
	B3	1051	Test condition G.
*	B3	2052	PIND, required if not performed in screening. (22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	B4	1042	Test condition D; the heating cycle shall be 1 minute minimum. No heat sink or forced air cooling on the device shall be permitted during the cycle.
	B5	1042	A separate sample may be pulled for each test. Accelerated steady-state reverse bias, test condition A, V_{DS} = rated, T_A = +175°C, t = 120 hours, read and record $V_{(BR)DSS}$ (pre and post at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and I_{DSS} shall not exceed 25 µA. Accelerated steady-state gate bias, condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
	B5	2037	Bond strength; test condition D.

4.4.2.2 Quality levels JAN, JANTX, and JANTXV.

Subgroup	Method	Condition
B2	1051	Test condition G.
B3	1042	Test condition D, 2,000 cycles minimum. The heating cycle shall be 1 minute minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows.

	Subgroup	Method	Condition
	C2	2036	Tension: Test condition A; weight = 10 lbs (4.54 kg), t = 10 s (not applicable to "U" suffix version).
*	C3	2052	PIND, required if not performed in screening. (JANS only, 22 devices, c = 0 for large lots, 12 devices, c = 0 for small lots).
	C5	3161	See 4.3.3, $R_{\theta JC(max)}$ shall be 1.0°C/W.
	C6	1042	Test condition D. The heating cycle shall be 1 minute minimum.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in MIL-STD-750.

4.6 Element evaluation of un-encapsulated die (quality levels JANHC and JANKC). The element evaluation of un-encapsulated die shall be in accordance with appendix G of MIL-PRF-19500.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.3	$Z_{\theta JC}$			°C/W
Breakdown voltage, drain to source	3407	Bias condition C, V_{GS} = 0 V, I_D = 1 mA dc	$V_{(BR)DSS}$			
2N7218, 2N7218U				100		V dc
2N7219, 2N7219U				200		V dc
2N7221, 2N7221U				400		V dc
2N7222, 2N7222U				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	Bias condition C, V_{GS} = 20 V dc, V_{DS} = 0	I_{GSSF1}		100	nA dc
Gate current	3411	Bias condition C, V_{GS} = −20 V dc, V_{DS} = 0	I_{GSSR1}		−100	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	Condition A, V_{GS} = 10 V dc, pulsed (see 4.5.1), I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
2N7218, 2N7218U					0.077	Ω
2N7219, 2N7219U					0.18	Ω
2N7221, 2N7221U					0.55	Ω
2N7222, 2N7222U					0.85	Ω
Static drain to source on-state resistance	3421	Condition A, V_{GS} = 10 V dc, pulsed (see 4.5.1), I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
2N7218, 2N7218U					0.125	Ω
2N7219, 2N7219U					0.25	Ω
2N7221, 2N7221U					0.70	Ω
2N7222, 2N7222U					0.95	Ω
Forward voltage (source drain diode)	4011	Condition A, V_{GS} = 0, I_D = rated I_{D1}, pulsed (see 4.5.1)	V_{SD}		1.5	V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 2 – Continued						
Forward voltage (source drain diode)	4011	Condition A, V_{GS} = 0, I_D = rated I_{D1}, pulsed (see 4.5.1)	V_{SD}		1.5	V
Subgroup 3						
High temperature operation:		$T_C = T_J$ = +125°C				
Gate current	3411	Bias condition C, V_{GS} = 20 V dc, V_{DS} = 0	I_{GSSF2}		200	nA dc
Gate current	3411	Bias condition C, V_{GS} = −20 V dc, V_{DS} = 0	I_{GSSR2}		−200	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0, V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
Drain current	3413	Bias condition C, V_{GS} = 0, V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA	$V_{GS(th)2}$	1.0		V dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, pulsed (see 4.5.1), I_D = rated I_{D2} (see 1.3)	$r_{DS(on)3}$			
2N7218, 2N7218U					0.24	Ω
2N7219, 2N7219U					0.48	Ω
2N7221, 2N7221U					1.44	Ω
2N7222, 2N7222U					2.04	Ω
Low temperature operation:		$T_C = T_J$ = −55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA	$V_{GS(th)3}$		5.0	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 4						
Switching time test	3472	I_D = rated I_{D2} (see 1.3), V_{GS} = 10 V dc, gate drive impedance = 9.1 Ω, V_{DD} = 50 percent of $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$			
2N7218, 2N7218U					21	ns
2N7219, 2N7219U					20	ns
2N7221, 2N7221U					25	ns
2N7222, 2N7222U					21	ns
Rise time			t_r			
2N7218, 2N7218U					105	ns
2N7219, 2N7219U					105	ns
2N7221, 2N7221U					92	ns
2N7222, 2N7222U					73	ns
Turn-off delay time			$t_{d(off)}$			
2N7218, 2N7218U					64	ns
2N7219, 2N7219U					58	ns
2N7221, 2N7221U					79	ns
2N7222, 2N7222U					72	ns
Fall time			t_f			
2N7218, 2N7218U					65	ns
2N7219, 2N7219U					67	ns
2N7221, 2N7221U					58	ns
2N7222, 2N7222U					51	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 6; tp = 10 ms, V_{DS} = 80 percent of rated $V_{BR(DSS)}$, V_{DS} = 200 V maximum				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7218, 2N7218U					59	nC
2N7219, 2N7219U					60	nC
2N7221, 2N7221U					65	nC
2N7222, 2N7222U					68.5	nC
Gate to source charge			Q_{gs}			
2N7218, 2N7218U					16	nC
2N7219, 2N7219U					14.6	nC
2N7221, 2N7221U					14.0	nC
2N7222, 2N7222U					12.5	nC
Gate to drain charge			Q_{gd}			
2N7218, 2N7218U					30.7	nC
2N7219, 2N7219U					37.6	nC
2N7221, 2N7221U					40.5	nC
2N7222, 2N7222U					42.4	nC
Reverse recovery time	3473	Condition A, di/dt ≤ 100 A/μs, V_{DD} ≤ 30 V, $I_D = I_{D1}$, (see 1.3)	t_{rr}			
2N7218, 2N7218U					400	ns
2N7219, 2N7219U					500	ns
2N7221, 2N7221U					600	ns
2N7222, 2N7222U					700	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ For end-point measurements, this test required for the following subgroups:
Group B, subgroups 3 and 4 (JANS).
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroups 2 and 6.
Group E, subgroup 1.

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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices
Temperature cycling	1051	−55°C to +150°C, 500 cycles	
Hermetic seal Fine leak Gross leak	1071	As applicable.	
Electrical measurements		See table I, subgroup 2	
Subgroup 2 1/			45 devices, c = 0
Steady-state reverse bias	1042	Condition A, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Steady-state gate bias	1042	Condition B, 1,000 hours	
Electrical measurements		See table I, subgroup 2	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500	
Subgroup 5 2/			3 devices, c = 0
Barometric pressure test 2N7221, 2N7221U 2N7222, 2N7222U	1001	Condition C, $V_{(ISO)} = V_{DS}$ V_{DS} = 400 V dc V_{DS} = 500 V dc	
Subgroup 10			22 devices, c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476	Test conditions shall be derived by the manufacturer	

See footnotes at end of table.

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TABLE II. Group E inspection (all quality levels) for qualification or re-qualification only - Continued.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 11 Repetitive avalanche energy	3469	Peak current $I_{AR} = I_D$; peak gate voltage V_{GS} = 10 V; gate to source resistor, R_{GS} 25 ≤ R_{GS} ≤ 200 Ω Temperature = T_J = +150°C +0, −10°C Inductance = $\left[\frac{2E_{AR}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum Number of pulses to be applied = 3.6 X 10^8, supply voltage (V_{DD}) = 50 V, time in avalanche = 2 µs minimum, 20 µs maximum. Frequency = 500 Hz minimum	5 devices, c = 0
Electrical measurements		See table I, subgroup 2	

1/ A separate sample for each test may be selected.
2/ Not required for 2N7218, 2N7218U, 2N7219, and 2N7219U.

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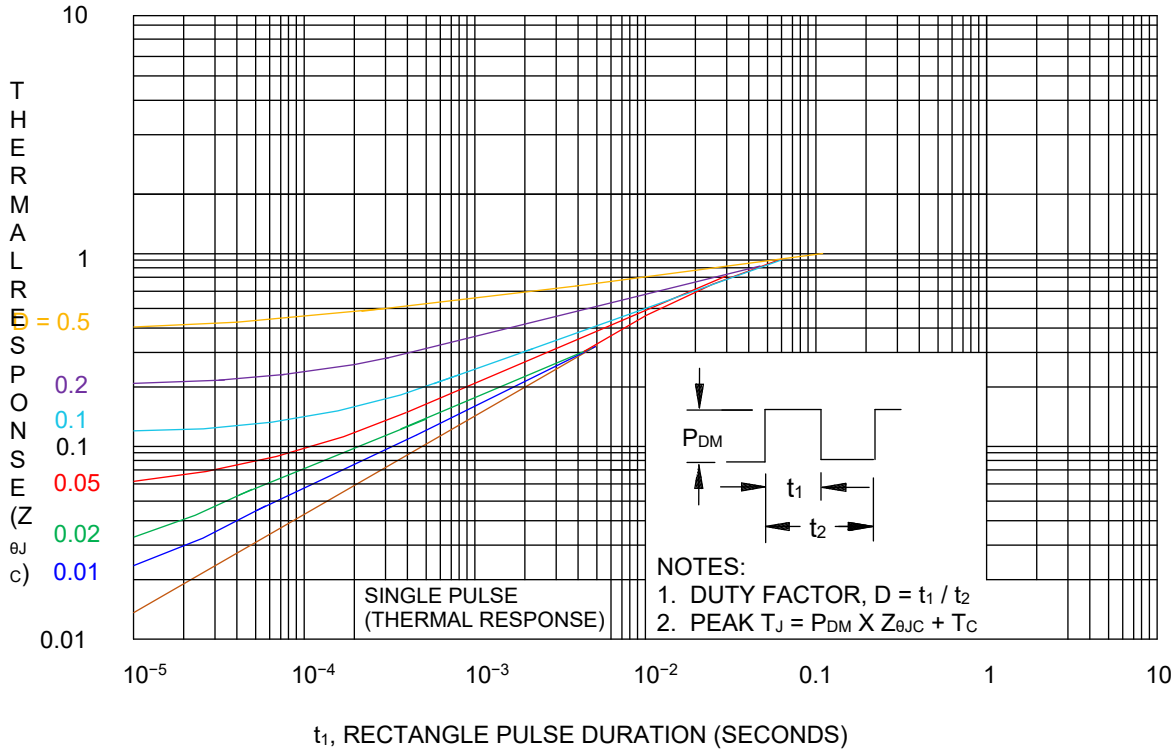


FIGURE 4. Thermal impedance.

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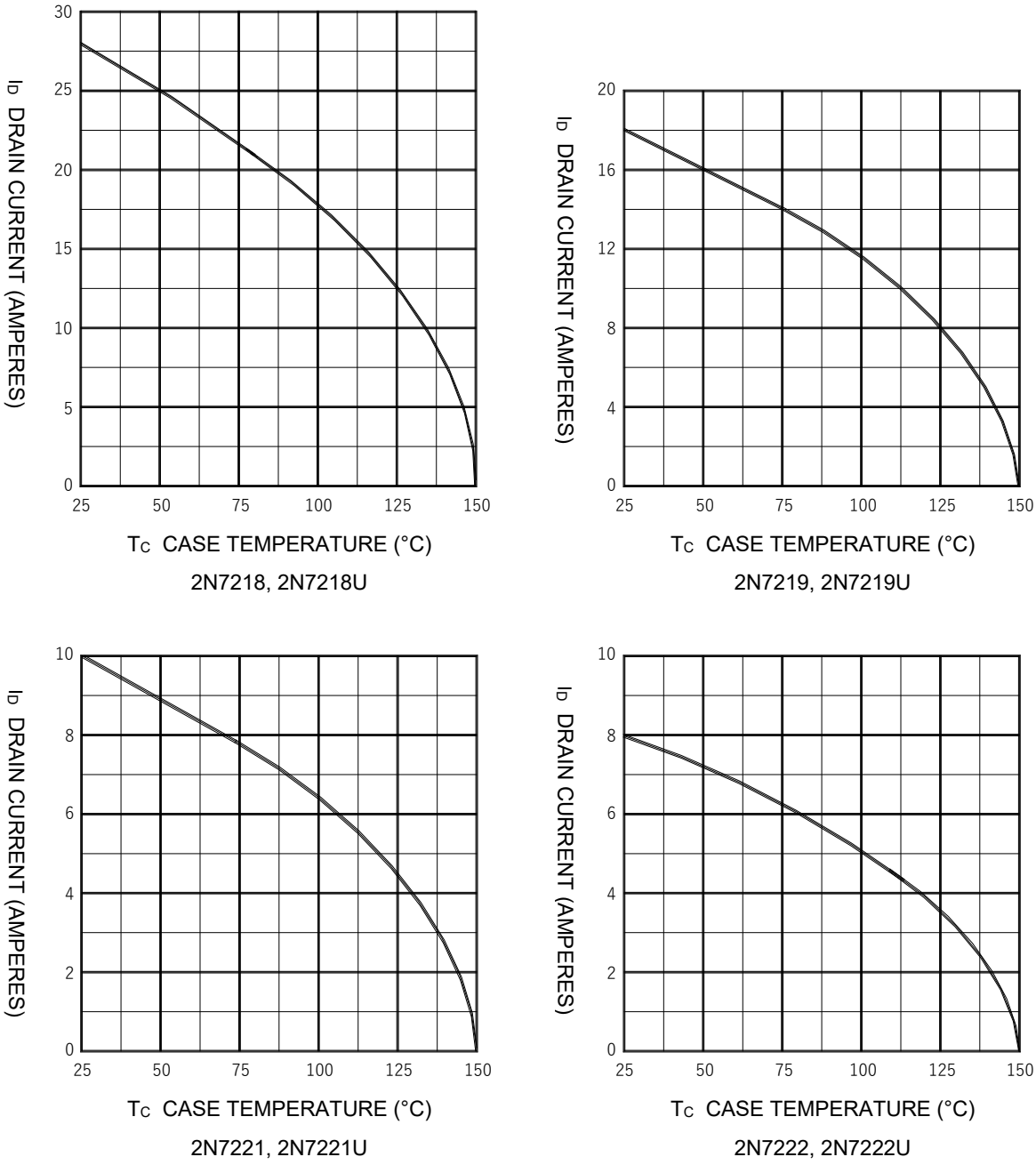


FIGURE 5. Maximum drain current versus case temperature graphs.

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2N7218, 2N7218U

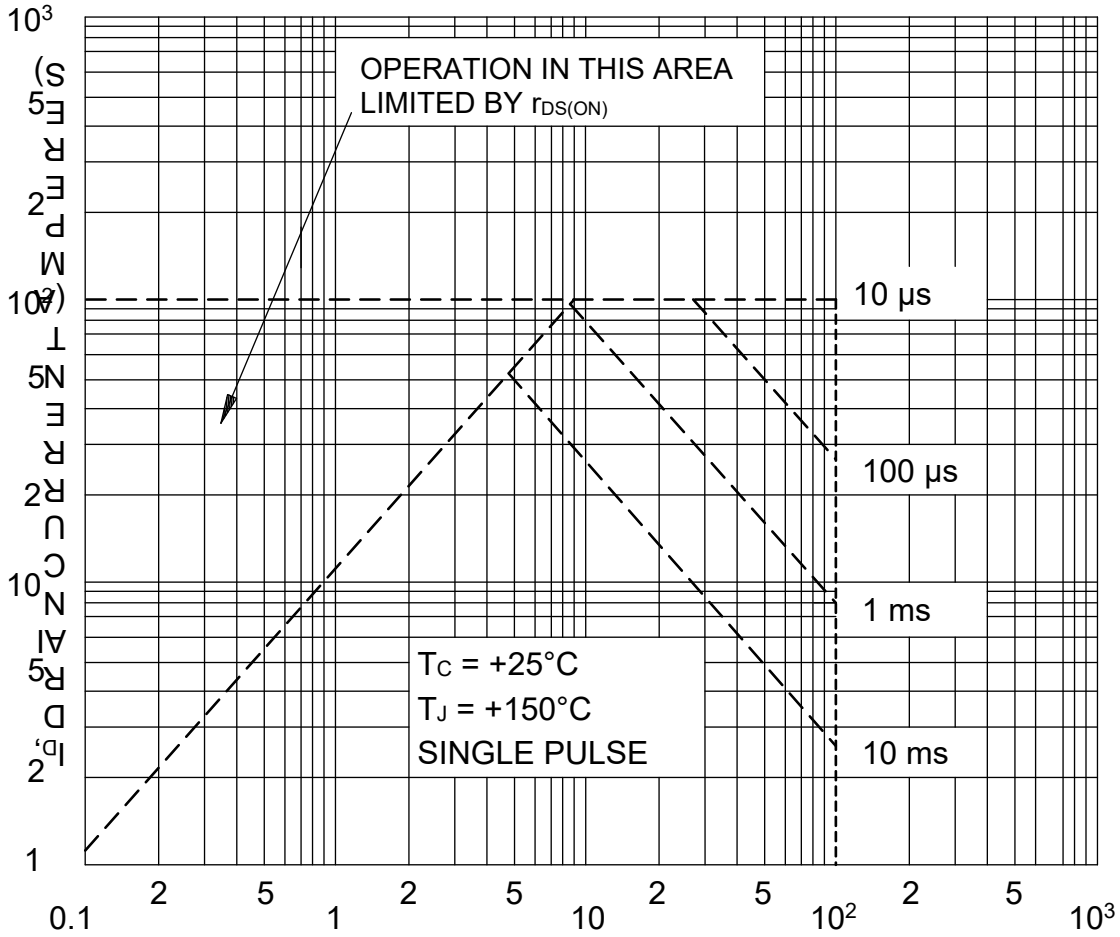
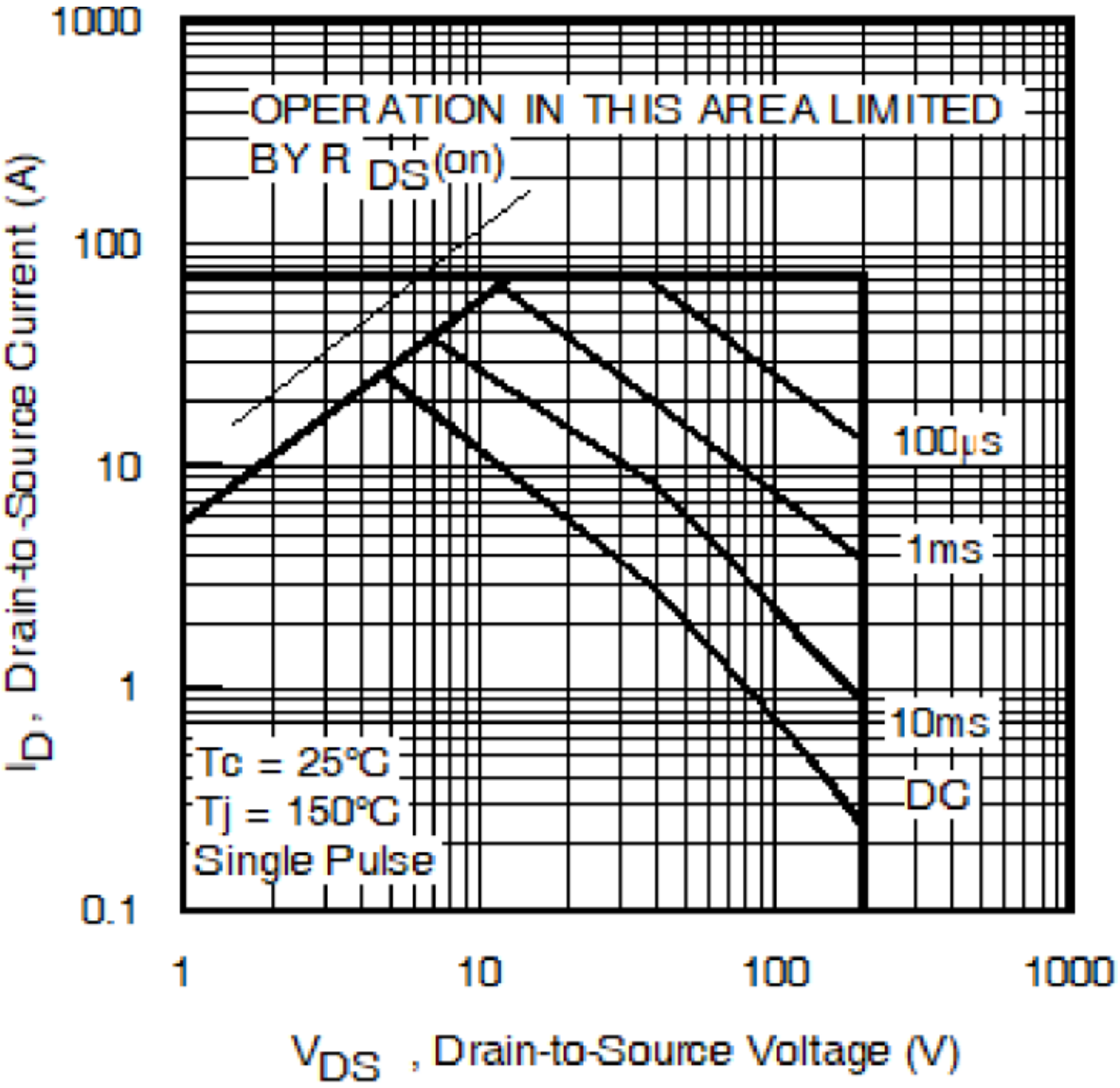


FIGURE 6. Safe operating area.

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* FIGURE 6. Safe operating area - Continued.

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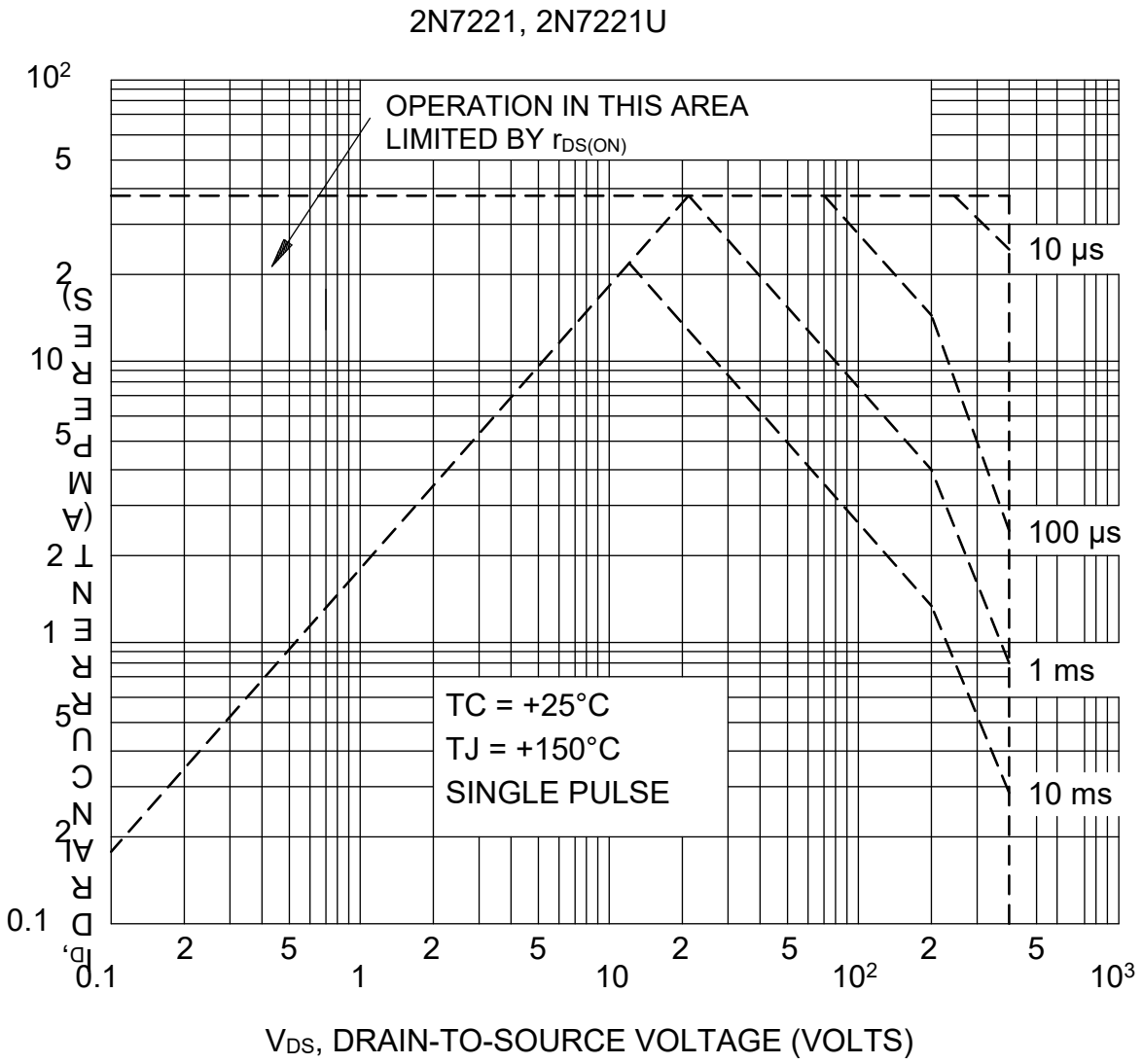


FIGURE 6. Safe operating area - Continued.

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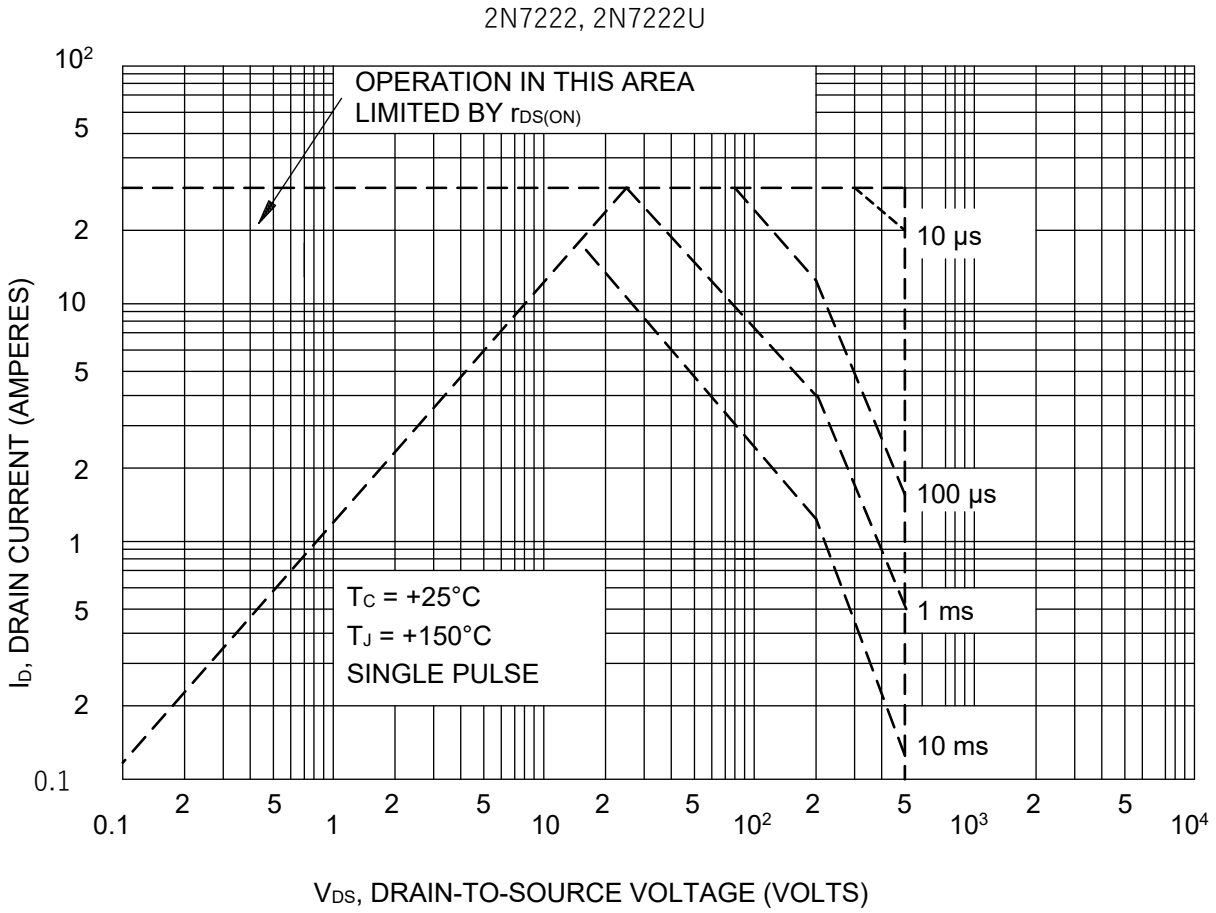


FIGURE 6. Safe operating area - Continued.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

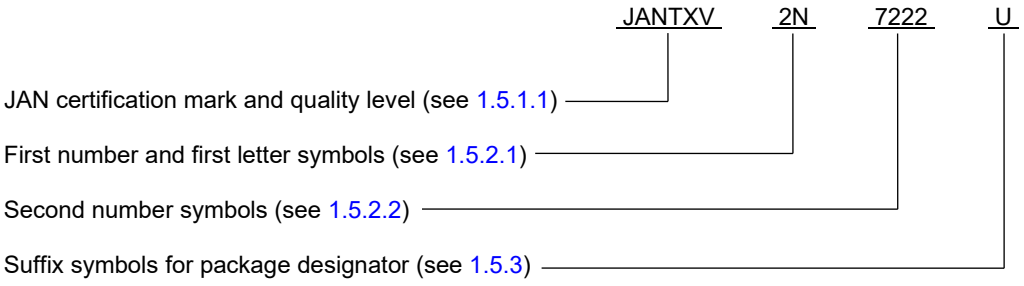
c. Lead formation and finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

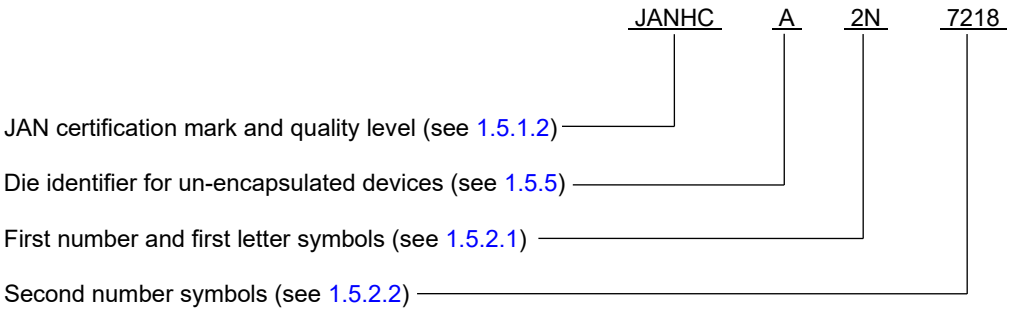
6.4 PIN construction examples.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



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6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
PINs for devices in a axial package			
JAN2N7218	JANTX2N7218	JANTXV2N7218	JANS2N7218
JAN2N7219	JANTX2N7219	JANTXV2N7219	JANS2N7219
JAN2N7221	JANTX2N7221	JANTXV2N7221	JANS2N7221
JAN2N7222	JANTX2N7222	JANTXV2N7222	JANS2N7222
PINs for devices in a SMD package			
JAN2N7218U	JANTX2N7218U	JANTXV2N7218U	JANS2N7218U
JAN2N7219U	JANTX2N7219U	JANTXV2N7219U	JANS2N7219U
JAN2N7221U	JANTX2N7221U	JANTXV2N7221U	JANS2N7221U
JAN2N7222U	JANTX2N7222U	JANTXV2N7222U	JANS2N7222U

6.5.2 List of PINs for un-encapsulated die. The following is a list of possible PINs for un-encapsulated die available on this specification sheet.

Quality level "HC"	Quality level "KC"
JANHCA2N7218	JANKCA2N7218
JANHCA2N7219	JANKCA2N7219
JANHCA2N7221	JANKCA2N7221
JANHCA2N7222	JANKCA2N7222

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6.6 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturer's CAGE	Manufacturer's and user's PIN
2N7218	59993	IRFM 140
2N7219	59993	IRFM 240
2N7221	59993	IRFM 340
2N7222	59993	IRFM 440
2N7218U	59993	IRFN 140
2N7219U	59993	IRFN 240
2N7221U	59993	IRFN 340
2N7222U	59993	IRFN 440

6.7 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example JANHCA2N7218) will be identified on the QML.

JANC ordering information		
Military PIN	Manufacturer	
	59993	59993
2N7218	JANHCA2N7218	JANKCA2N7218
2N7219	JANHCA2N7219	JANKCA2N7219
2N7221	JANHCA2N7221	JANKCA2N7221
2N7222	JANHCA2N7222	JANKCA2N7222

6.8 Request for new types and configurations. Requests for new device types or configurations for inclusions in this specification sheet should be submitted to: DLA Land and Maritime, ATTN: VAC, Post Office Box 3990, Columbus, OH 43218-3990 or by electronic mail at Semiconductor@dla.mil or by facsimile (614) 693−1642 or DSN 850−6939.

* 6.9 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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